

CUSTOMER SEGMENTATION ANALYSIS

Name: Samrendra Narayan Singh

Student ID: 47961171

Word Count: 914 words

Introduction and Exploratory Data Analysis

Based on purchasing behavior and demographics data, this customer segment analysis will classify 2000 customers. To improve engagement and sales, the aim is to create specific/ targeted marketing strategies for the travel agency.

The process involves various stages including exploratory data analysis, pre-processing of the data, and clustering techniques (K means ++ and Agglomerative clustering in this scenario) to identify unique customer segments that have not been tapped into and can be explored further.

Gender, Marital Status, Age, Education, Income, Occupation, and Settlement Size are the variables included in the data.

	Gender	Marital Status	Age	Education	Income	Occupation	Settlement Size
count	2000.00000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000
mean	0.60450	0.500500	40.823500	1.456500	137516.196500	0.612500	0.834000
std	0.48908	0.500125	9.455848	0.783846	46184.296588	0.674219	0.967942
min	0.00000	0.000000	20.000000	0.000000	35832.000000	0.000000	0.000000
25%	0.00000	0.000000	33.000000	1.000000	101262.750000	0.000000	0.000000
50%	1.00000	1.000000	40.000000	1.000000	133004.000000	1.000000	0.000000
75%	1.00000	1.000000	48.000000	2.000000	171232.500000	1.000000	2.000000
max	1.00000	1.000000	76.000000	3.000000	309364.000000	2.000000	2.000000

Gender is skewed with more females compared to males (females = 60%, males = 40%).

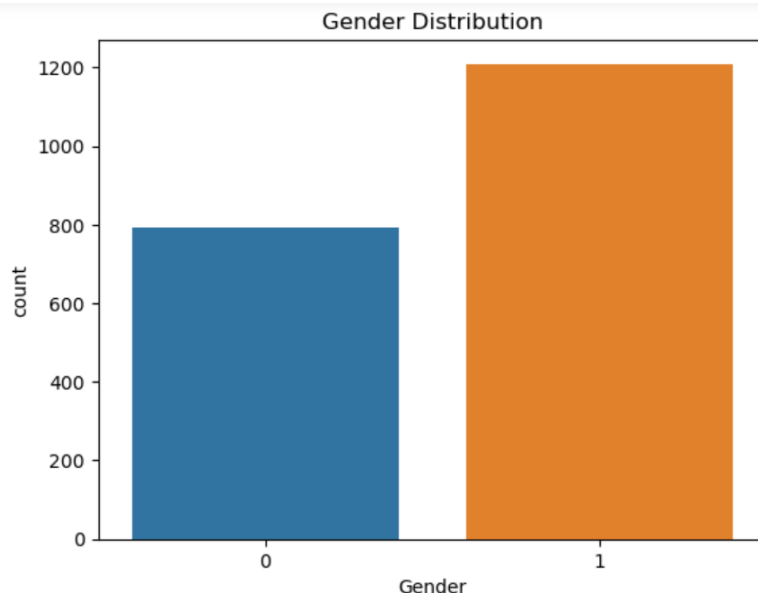
Marital status is well-balanced with the mean = 0.500500.

Age varies between 20 years (minimum), and 76 years (maximum) and the mean age is around 41 years.

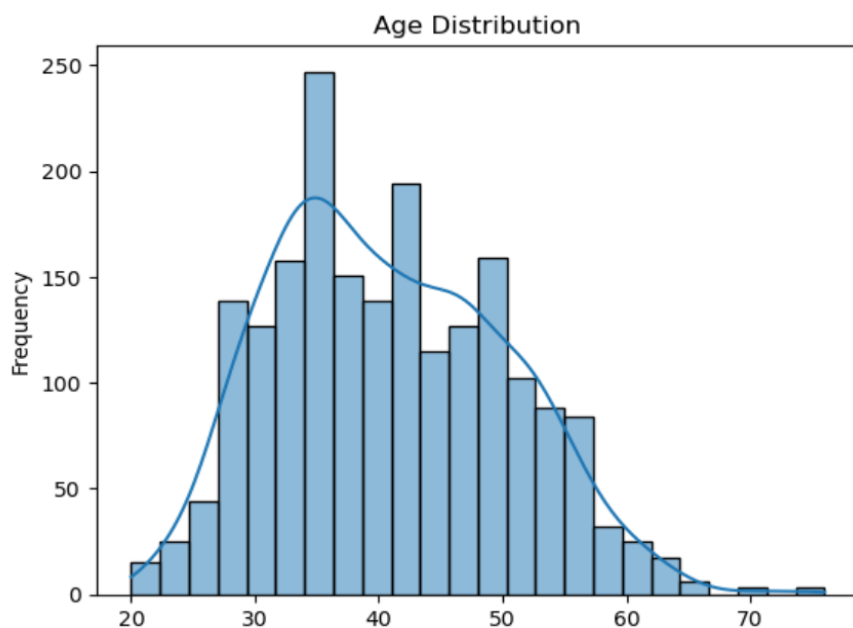
Similarly, income is in the range between USD 35832 (minimum), and USD 309364 (maximum) with the average income being USD 137516.1965.

Education, occupation, and settlement size display distributions varying over a big range.

These numbers tell us about the plethora of observations in the dataset, highlighting its wide range and intriguing us to find insights by deeply analyzing the data.



The above bar plot represents genders (males and females) and their frequencies in the dataset. Females are represented by the number one and males are represented by the number zero. Based on the above data and its visual representation, we can conclude that the gender data is skewed towards females, as females are greater than males in terms of frequency (females constitute roughly sixty percent, whereas males make up roughly forty percent of the gender demographic).



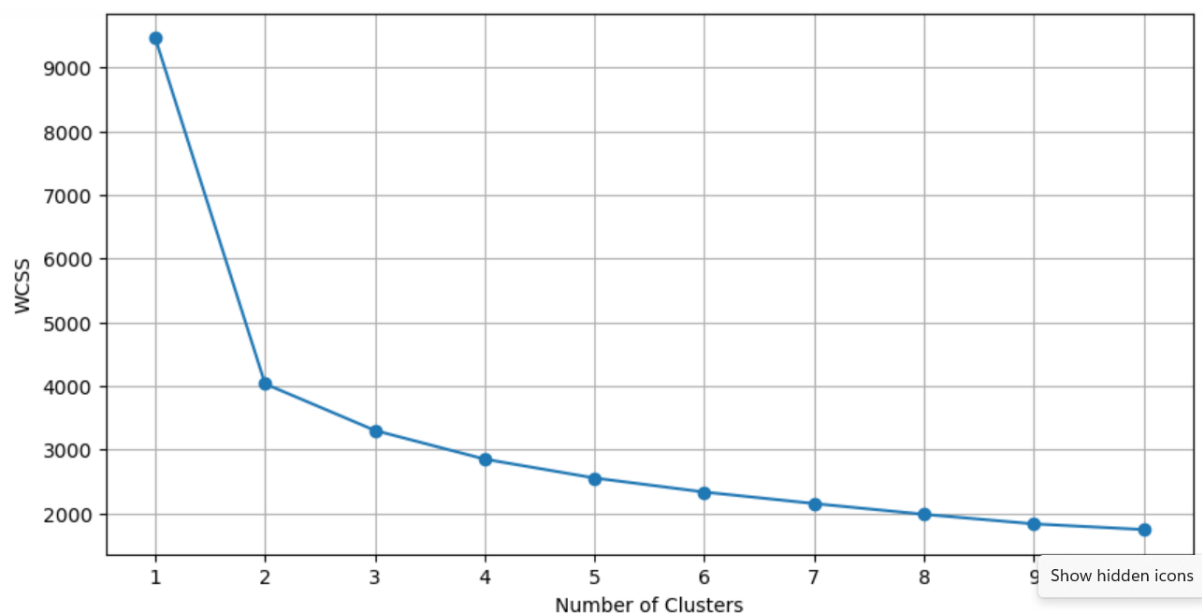
The above plot represents the age distribution of the various customers. The distribution is skewed towards the right.

It covers a wide range of ages, providing a complete and broad picture representation. Frequencies peak around the age of 35 years (Meaning mode is approximately 35 years).

The minimum age is 20 years, the maximum age is 76 years and the mean age is 41 years.

The median age is 40 years, which means that 50% of customers are less than 40 years old and the other half is older than 40 years.

Customer Segmentation



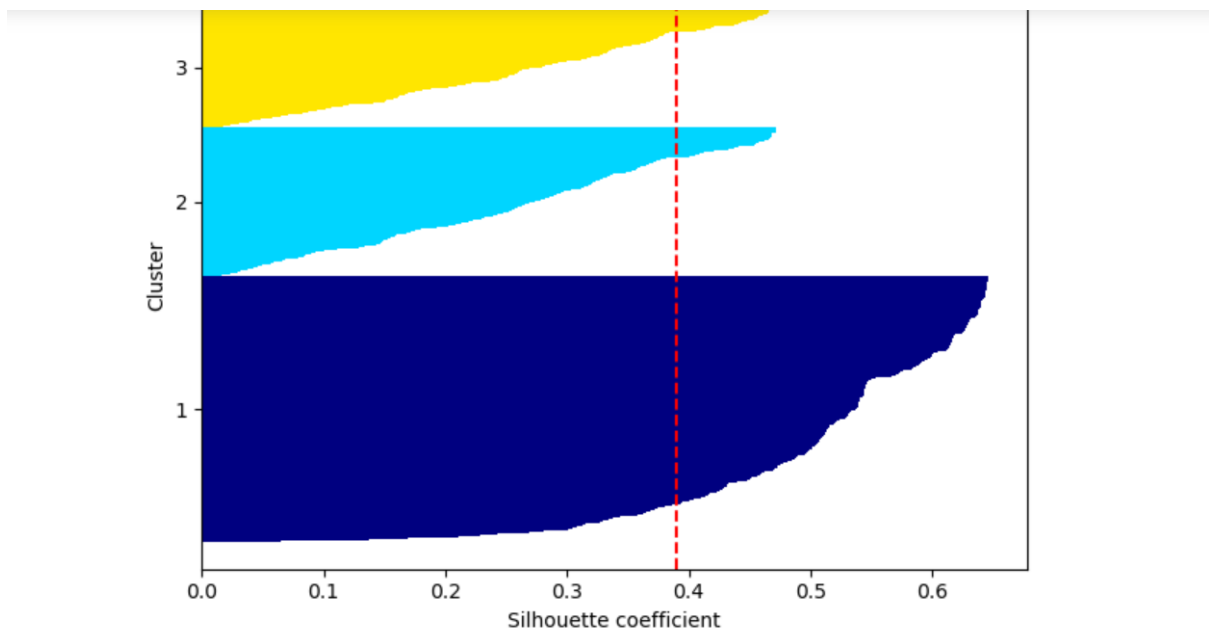
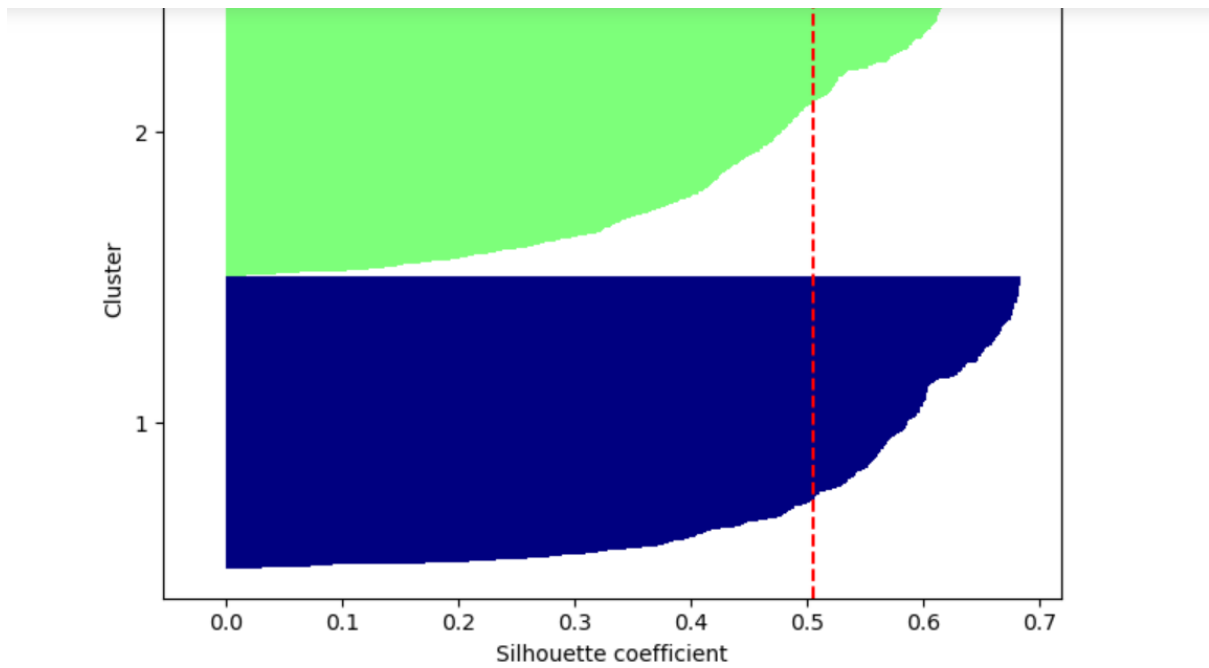
The above elbow plot represents the intricate relationship between the Number of Clusters and the Within Cluster Sum of Squares (WCSS). X-axis represents the Number of Clusters and Y axis represents Within Cluster Sum of Squares

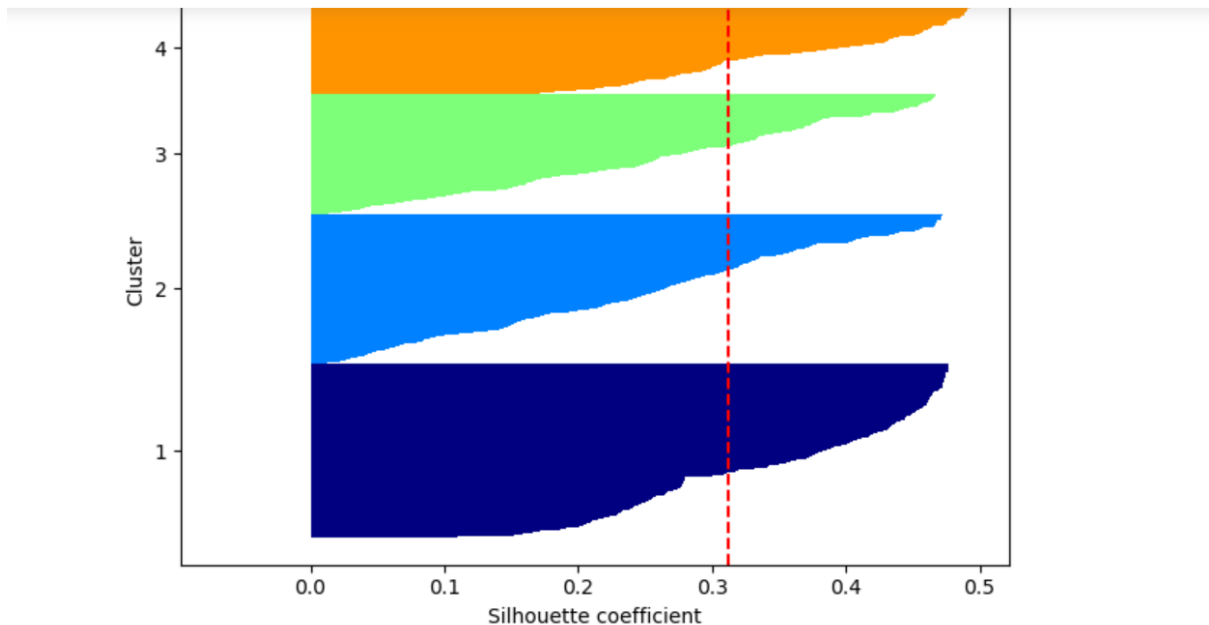
The elbow plot helps determine the optimal number of clustering for the K Means ++ clustering method. We can observe that with the increasing number of clusters, the Within Cluster Sum of Squares (WCSS) gradually declines, forming a bend known as the elbow. Hence the name is elbow plot.

This elbow which can be seen between 2 to 3 clusters, tells about striking the right balance between declining variance and avoiding overfitting.

After this point, we can see diminishing returns upon the addition of more clusters.

Silhouette Analysis





Silhouette coefficient varies between -1 to 1 . Higher values indicate good, well-defined clusters. Negative values tell about misclassified data points. Scores near zero suggest overlapping clusters. Scores close to one indicate distinct clusters, properly separated from others.

Silhouette analysis involves measuring intra-cluster cohesion and inter-cluster separation to assess clustering quality.

The average silhouette scores for the 2, 3, and 4 clusters are 0.51, 0.39 and 0.31 respectively. The score of 0.51 for 2 clusters suggests the best defined structure, meaning 2 is the optimal number of clusters in the given scenario.

KMeans ++ Clustering

```
[1 0 0 ... 1 0 1]
```

Clusters:

```
[ [-7.41890526e-01 -7.42425398e-01 6.53266332e-01 3.46733668e-01
 1.00000000e+00 -9.99200722e-16 1.91959799e-01 8.08040201e-01
-6.10622664e-16 1.11022302e-16 9.36683417e-01 6.23115578e-02
 1.00502513e-03 9.92964824e-01 7.03517588e-03 1.80411242e-16]
 [ 1.93332206e-01 1.93272864e-01 1.37745975e-01 8.62254025e-01
-2.22044605e-16 1.00000000e+00 1.78890877e-03 6.97674419e-02
 7.51341682e-01 1.77101968e-01 1.98568873e-01 1.78890877e-02
 7.83542039e-01 1.78890877e-03 7.79964222e-01 2.18246869e-01]
 [ 1.41279904e+00 1.41406668e+00 1.43497758e-01 8.56502242e-01
 8.96860987e-03 9.91031390e-01 -5.55111512e-17 7.62331839e-02
 7.55605381e-01 1.68161435e-01 1.95067265e-01 -2.08166817e-17
 8.04932735e-01 6.72645740e-03 7.80269058e-01 2.13004484e-01]]
```

The model runs 10 times with different random centroids to ensure robustness, iterating up to 300 times for convergence. Random state = 0 ensures reproducibility.

`y_km_plus` is an array indicating cluster assignments for each data point in `x`. The first point is assigned to cluster 1, the second to cluster 0, and so on.

The `km_plus.cluster_centers_` array displays the centroids of the clusters, each represented by 16 feature values (coordinates in the feature space). These centroids are the mean positions of all points in their respective clusters, defining the center of each cluster

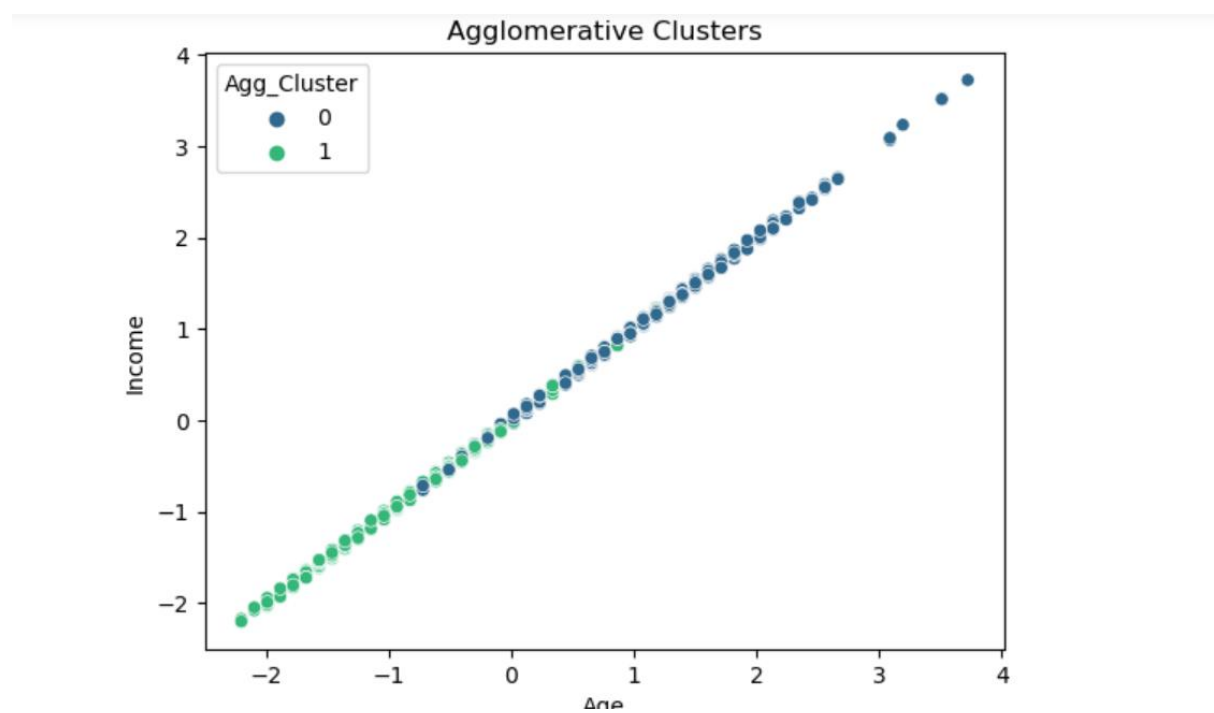
Agglomerative Clustering

```
Cluster labels: [0 0 0 ... 0 0 0]
```

The clustering is performed using the Euclidean distance metric and the 'complete' linkage method, which considers the maximum distance between points in different clusters while merging them. The `fit_predict` method returns the cluster labels for each data point, stored in the `labels` array.

The output cluster labels indicates the cluster assignment for each data point

Visualization of Clusters



Recommendations and Conclusion

	Age	Income	Gender_0	Gender_1	Marital Status_0	\
KMeans_Cluster						
0	0.753173	0.753792	0.139980	0.860020	0.004028	
1	-0.742702	-0.743312	0.647468	0.352532	0.988083	
	Marital Status_1	Education_0	Education_1	Education_2	\	
KMeans_Cluster						
0	0.995972	0.001007	0.070493	0.756294		
1	0.011917	0.189672	0.801390	0.005958		
	Education_3	Settlement Size_0	Settlement Size_1	\		
KMeans_Cluster						
0	0.172205	0.194361	0.004028			
1	0.002979	0.930487	0.067527			
	Settlement Size_2	Occupation_0	Occupation_1	Occupation_2	\	
KMeans_Cluster						
0	0.801611	0.004028	0.780463	0.215509		
1	0.001986	0.981132	0.015889	0.002979		
	Agg_Cluster					
KMeans_Cluster						
0	0.004028					
1	0.987090					

The clusters represent distinct/ unique groups based on standardized income, standardized age, and categorical variables

Cluster 0: Higher age and income. Mostly females, non-single, educated till university level, and living in a big city. Most of them are skilled employees/officials with some of them in management/self-employed/ highly skilled employees/officers.

Recommendations:

The travel agency should target high-end, expensive, and exclusive destinations focusing on urban outreach and marketing campaigns.

Cluster 1: Lower age and income. Mostly males, single, maximum education level till high school, mostly living in a mid-sized city or small city. Most of them are unskilled/ unemployed, with a minuscule number being skilled/ employed.

Recommendations:

The travel agency should target less expensive and more affordable destinations, focusing on rural outreach and marketing campaigns.

Conclusion

The above analysis helps us identify distinct (unique) socio-economic and demographic segments, making it easy to tailor marketing strategies.

This approach will enhance customer satisfaction and business efficiency, addressing the needs of each cluster.